Indian Institute of Technology, Kharagpur

AUTOMATED DETECTION OF MALARIA-INFECTED CELLS USING CONVOLUTIONAL NEURAL NETWORKS

Presented by: Raghav Agarwal (21BT3EP20)
Dual Degree (B.Tech + M.Tech) in Biotechnology & Biochemical Engineering/Engineering Entrepreneurship

Supervisor: Prof. Shailendra Kumar Varshney

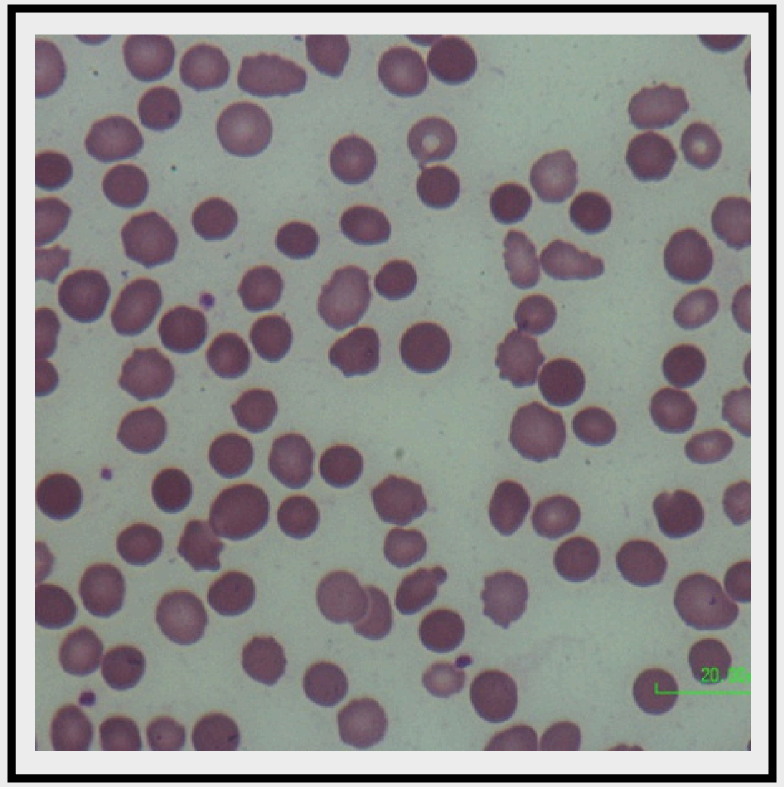
INTRODUCTION

- Context:
  - Malaria affects 200+ million people globally every year, with high mortality. Caused by protozoan parasites of the Plasmodium genus
  - Early diagnosis is vital for effective treatment and reducing transmission.
- Challenges in Traditional Diagnosis:
  - Reliance on expert microscopists.
  - Time-intensive and prone to human error.
  - Limited accessibility in resource-constrained areas.
- Proposed Solution:
  - Automation through machine learning and image analysis.
  - Using Convolutional Neural Networks (CNNs) to classify red blood cells (RBCs) efficiently.

PROJECT OBJECTIVES

- Primary Goals:
  - Develop a reliable, automated pipeline for malaria diagnosis.
  - Integrate image segmentation and CNN-based classification.
- Specific Objectives:
  - Accurately isolate RBCs from high-resolution blood smear images.
  - Differentiate malaria-infected and non-infected cells with high sensitivity and specificity.
- Broader Impact:
  - Improve diagnostic capabilities in under-resourced settings.
  - Lay the groundwork for future advancements in automated pathology.

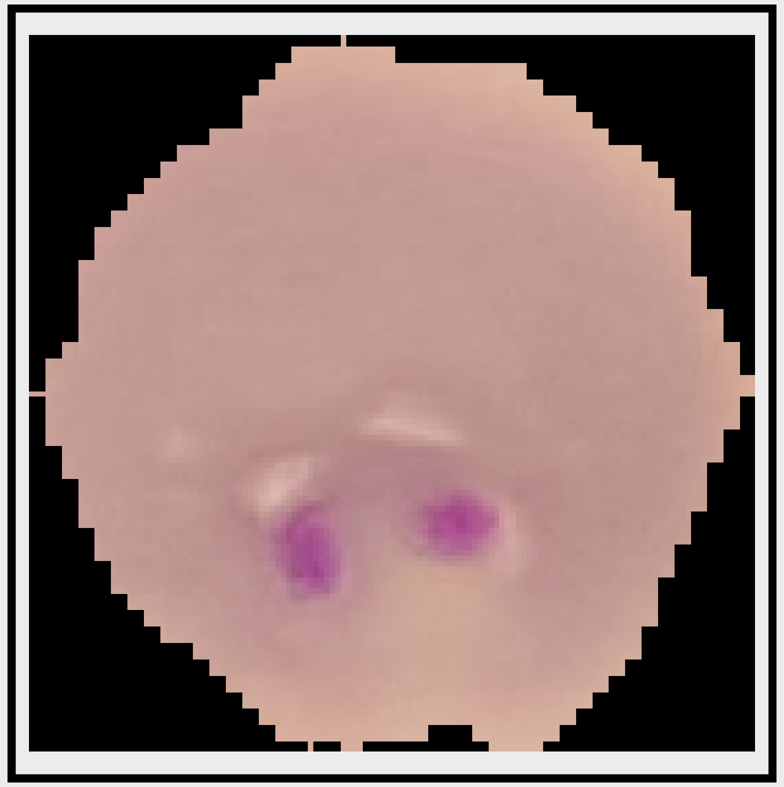
DATASET AND EXPERIMENTAL SETUP



Segmentation Task:

- Dataset: 706 high-resolution blood smear images (20,000+ labeled RBCs in 12 classes).
- Preprocessing: Resizing (512x512), random rotations, flips, and intensity normalization.
- Model: U-Net with encoder-decoder architecture and skip connections.
- Training: 100 epochs with Dice loss and Adam optimizer (LR: 0.0005).

DATASET AND EXPERIMENTAL SETUP



Classification Task:

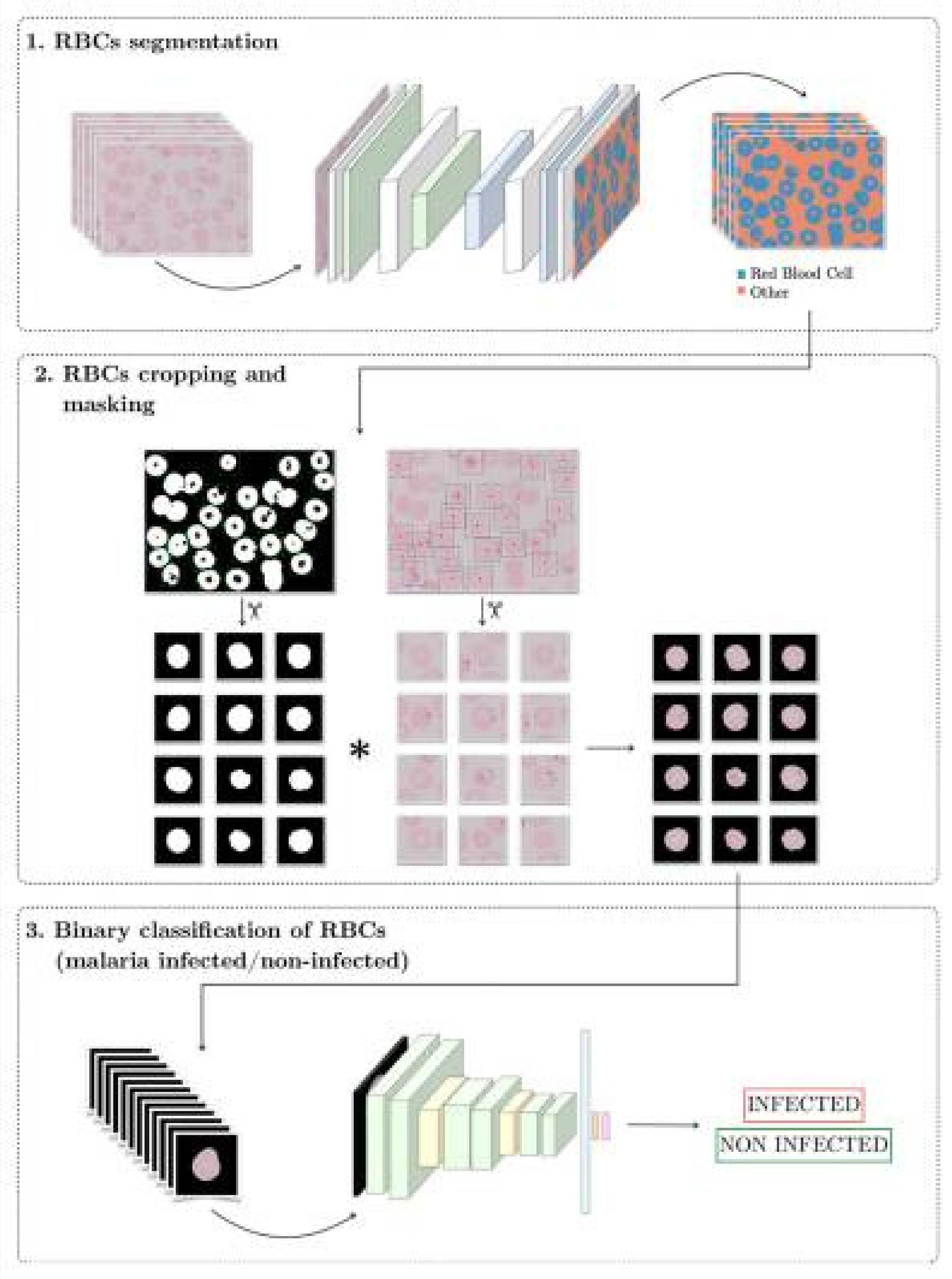
- Dataset: National Library of Medicine (NLM) malaria dataset.
- Preprocessing: Normalization, resizing (128x128), and augmentation.
- Model: Custom CNN with binary classification.
- Training: 75 epochs with Adam optimizer and binary cross-entropy loss.

METHODOLOGY

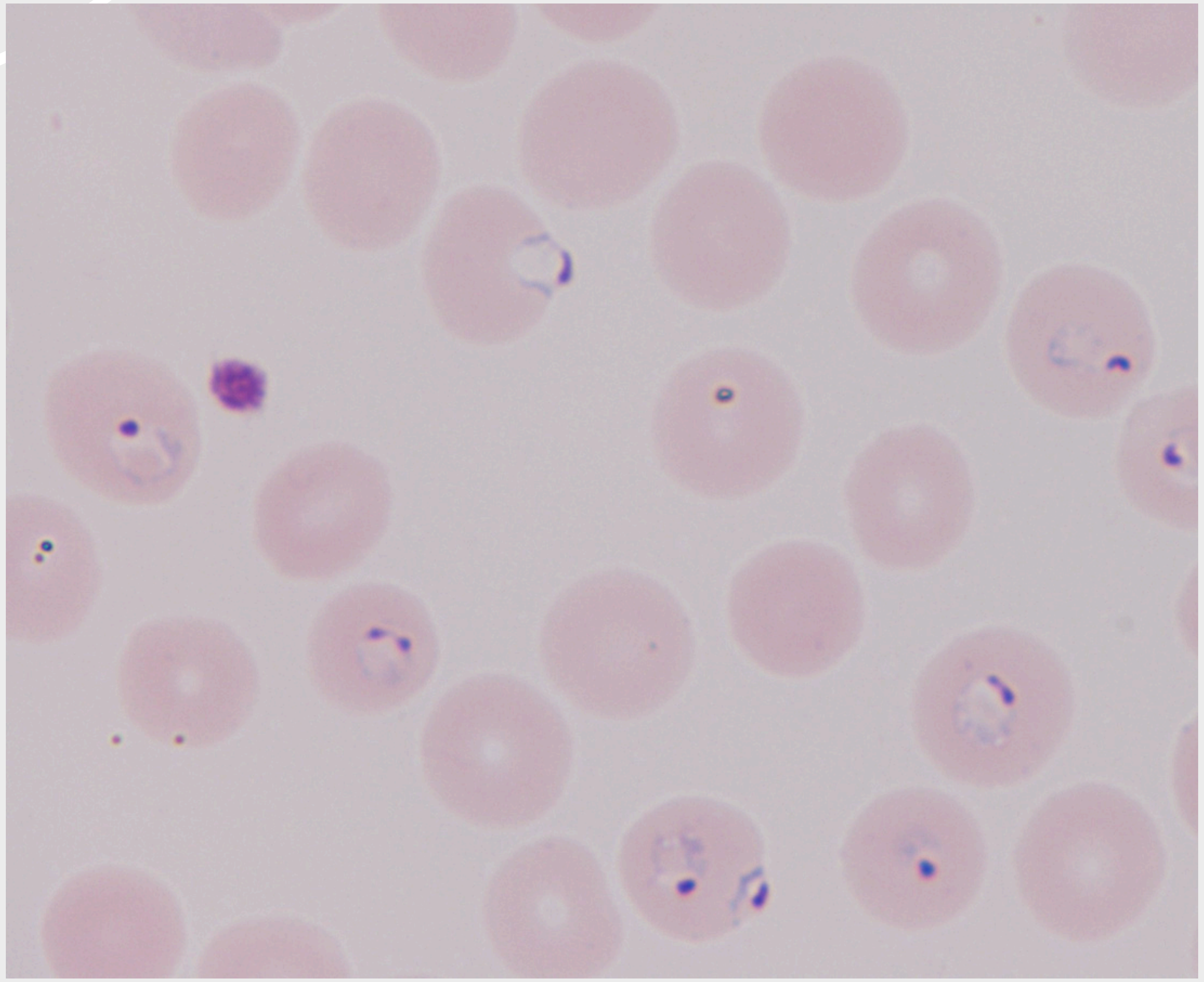
Pipeline Overview:

1. Input: High-resolution blood smear images.
2. Preprocessing: Image resizing, normalization, and augmentation to improve model generalization.
3. Segmentation: U-Net isolates individual RBCs for analysis.
4. Classification: CNN labels segmented cells as malaria-infected or non-infected.

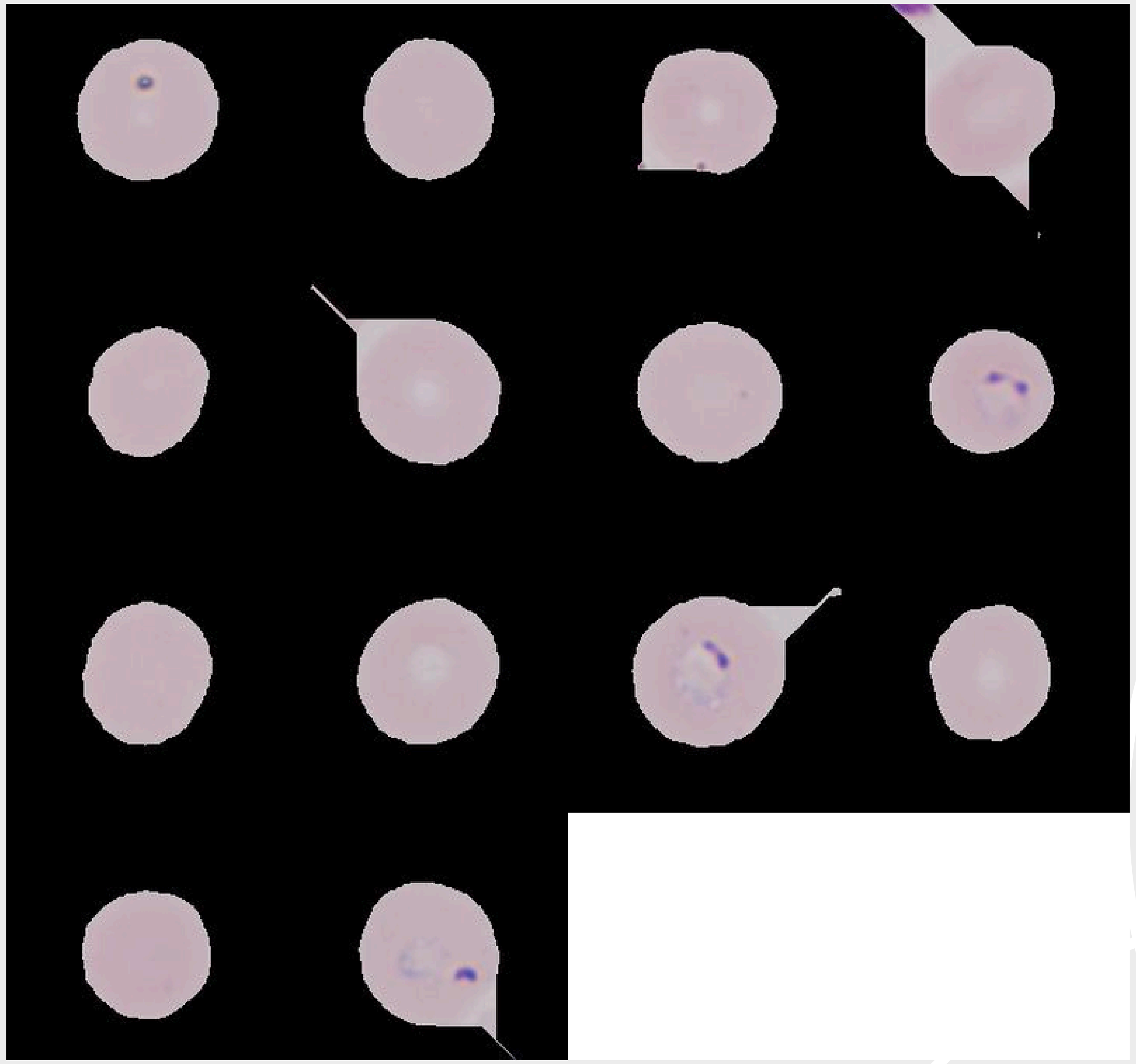
Flowchart Illustration on next page.....


1. RBCs segmentation
Red Blood Cell
Other
2. RBCs cropping and masking
3. Binary classification of RBCs (malaria infected/non-infected)
INFECTED
NON INFECTED

RBC segmentation



Sample Blood Smear Image

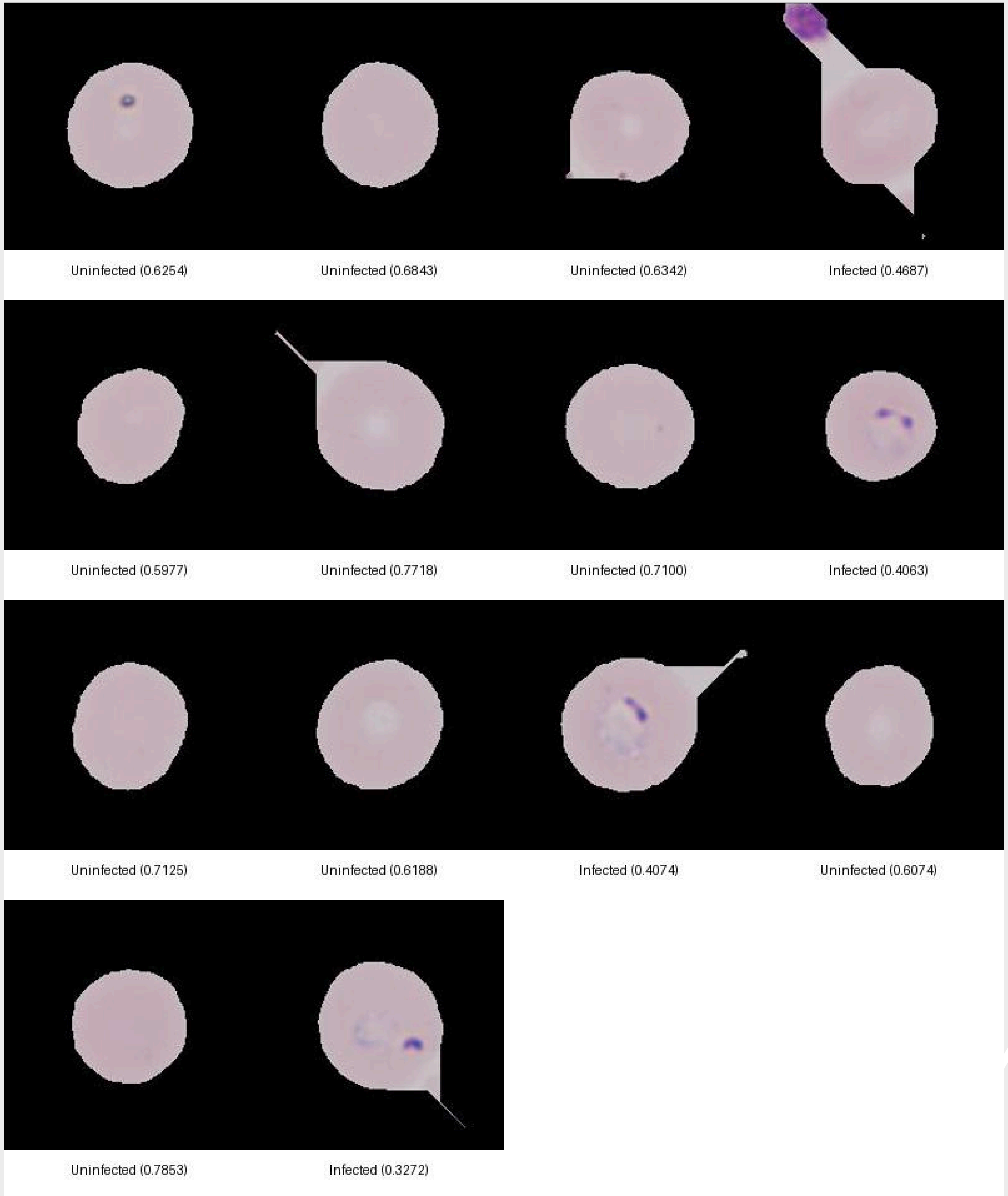


Segmented RBCs

Classified RBCs

Patient data from AIIMS Gorakhpur

Sample Result log



RESULT

- Performance Segmentation:
  - Accuracy: 93.72%.
  - Jaccard similarity coefficient: 0.805–0.947 (high agreement with ground truth).
  - Effective segmentation of overlapping or clustered RBCs.
- Performance Classification:
  - Validation accuracy: 95%.
  - Test accuracy: 75.39% (drop due to dataset imbalance).
  - Sensitivity: 17.9% (many infected RBCs missed).
  - Specificity: 87.04% (accurate identification of non-infected RBCs).

CHALLENGES

- Misclassification of central RBC regions and platelets as background.
- False positives due to resemblance between platelets and RBCs.
- High false positives due to misclassification of platelets.

CONCLUSION

KEY FINDINGS

- Successfully integrated segmentation (U-Net) and classification (CNN) for automated malaria detection.
- Segmentation performed well, while classification showed room for improvement in sensitivity.

FUTURE WORK

- Incorporate Indian cell data into the dataset to enhance its diversity.
- Test robustness in diverse clinical settings with real-world blood smears.

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THANK YOU

Presented by Raghav Agarwal